

Emmanuel Tega Agbragu

Application Security Engineer

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SUMMARY

Application Security Engineer with 4+ years of experience across offensive and defensive security, specializing in manual web and API penetration testing, secure development lifecycle (SDLC) integration, and vulnerability management. Proven track record of identifying critical vulnerabilities and deploying enterprise security infrastructure for regulated fintech and banking institutions, with a strong focus on OWASP, NIST, and PCI DSS compliance.

EXPERIENCE

Security Researcher

Jan 2025 – Apr 2026

The Upanzi Network, Carnegie Mellon University | Kigali, Rwanda

- Conducted web and API penetration tests across research and application platforms, uncovering and exploiting vulnerabilities to deliver reproducible proof-of-concept evidence for remediation.
- Chained a mass-assignment flaw (privilege escalation) with a BOLA/IDOR vulnerability rated **Critical (CVSS 9.1)** to demonstrate full unauthorized CRUD operations on an e-commerce API, quantifying business impact for prioritization.
- Identified an indirect prompt-injection vulnerability (OWASP **LLM01**) in an LLM-powered dataset feature, proving model-behavior manipulation and surfacing AI-specific risks in the application security posture.
- Authored OWASP- and NIST-aligned VAPT reports for each engagement and drove the vulnerability management lifecycle through to verified closure by collaborating with development teams.
- Built custom Python automation scripts and extended mobile vulnerability scanners to scale endpoint enumeration and security testing across large asset inventories.

Graduate Teaching Assistant

Sep 2024 – Jan 2025

Carnegie Mellon University Africa | Kigali, Rwanda

- Delivered lab sessions in Ethical Hacking and Cybersecurity Operations, reinforcing student competency in offensive security, penetration testing, and defensive security solutions.

Cybersecurity Engineer

Apr 2021 – Jun 2023

Lumenave International Limited | Lagos, Nigeria

- Designed and deployed security infrastructure including IDS/IPS, EDR, and firewalls for banking clients, reducing malware infections by **40%** and enhancing IT security posture.
- Implemented CyberArk PAM solutions with role-based access controls and least-privilege policies to meet fintech compliance requirements and ISO 27001 standards.
- Configured Symantec DLP policies defining detection rules and access controls to prevent unauthorized disclosure and exfiltration of sensitive financial data.
- Led a team of 5 engineers delivering security projects on time and under budget, ensuring operational excellence for all deployed security solutions.
- Provided technical support and incident response assistance for deployed security solutions, maintaining **100%** operational efficiency for financial sector clients.

SELECTED PROJECTS

Securing the Digital State: Research Report

2025

Python, Automation | Upanzi Network

- Contributed to a pan-African study assessing the security posture of **21,782** government sub-domains across 54 countries.
- Led HTTP-methods enumeration and attack surface analysis, identifying dangerous exposed methods.
- Built custom Python automation scripts powering mass asset discovery and vulnerability scanning.

Privileged Access Management Deployment

2021 – 2023

CyberArk, Bash, Oracle, PowerShell | First City Monument Bank

- Deployed a complete CyberArk stack managing privileged account access across network perimeters.
- Implemented secure authentication and least-privilege role-based access controls aligned to PCI DSS and regulatory requirements.

TECHNICAL SKILLS

Security Tools	Burp Suite, OWASP ZAP, Nmap, Nessus, OpenVAS, SQLMap, Symantec DLP
Web & Networking	HTTP, TCP/IP, REST APIs, authentication (SAML, OAuth, LDAP, Kerberos), cloud security (AWS)
Programming	Python, Bash, C/C++, Java, SQL, PHP, JavaScript
Frameworks	OWASP Top 10, MITRE ATT&CK, NIST, ISO 27001, PCI DSS, SAST, DAST

EDUCATION

MSc Information Technology

Aug 2023 – May 2025

Carnegie Mellon University | GPA: 3.80/4.0 | Mastercard Foundation Partial Scholarship

BSc Pure and Applied Physics, First Class Honours

Mar 2014 – May 2018

University of Benin | Nigeria | GPA: 4.54/5.0 | Top 1% of graduating class

CERTIFICATIONS AND ACHIEVEMENTS

CEH Master • eJPT • AWS Certified Cloud Practitioner • OSCP (In Progress) • Top 10 Finalist, Greenfist CTF 2023 • Mastercard Foundation Scholarship 2023 • First Class Honours, Top 1%, 2018